

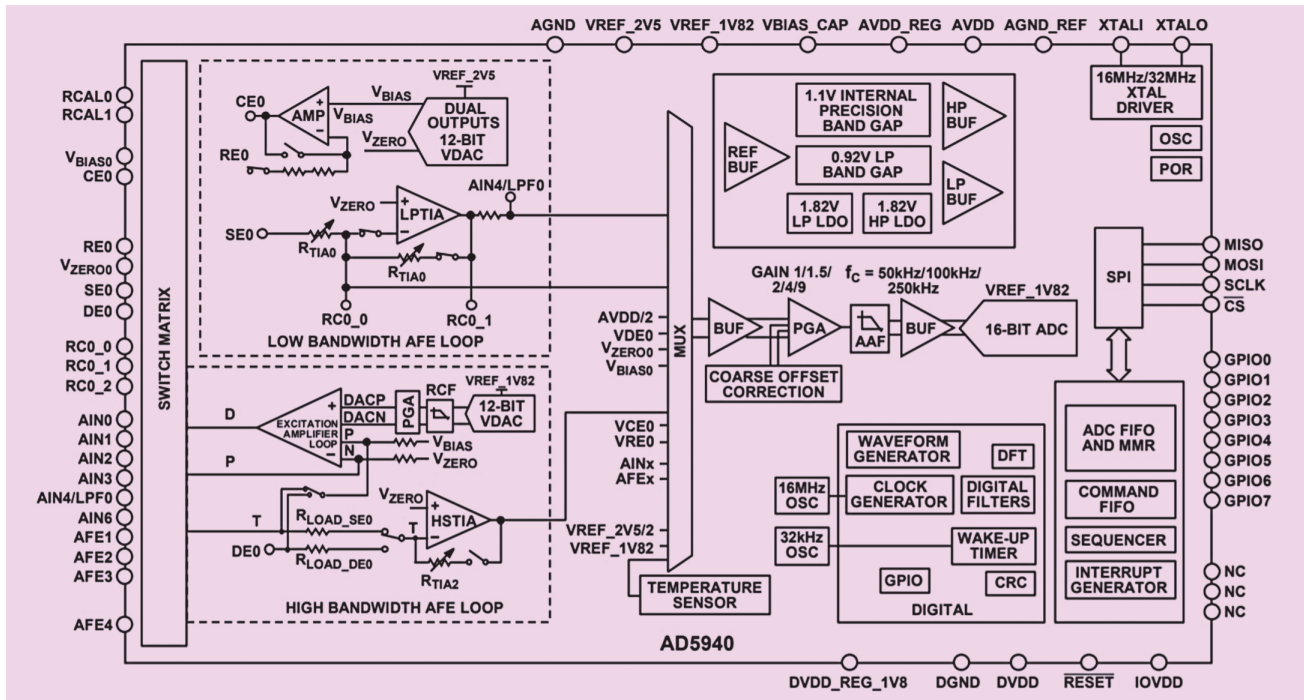


Fig. 6: The functional block diagram of AD5940 (Credit: Analogue Devices)

ent light and small movements of the sensor. If the heart rate signal is relatively small, like a signal from the human wrist, additional software signal processing might be required.

This proposed design, as seen in Fig. 4, is divided into eight major blocks:

- LED driver
- Transimpedance amplifier (TIA)
- Sample and hold circuit
- High-pass filter
- Offset correction circuit
- Biased non-inverting amplifier
- Analogue comparator with digital filter
- Analogue power supply filter and grounding considerations

The waveforms of the hardware prototype testing are shown in Fig. 5.

AD5940 and AD5941 by Analogue Devices. The AD5940 and AD5941 are high-precision, low power analogue front ends (AFEs), designed for skin impedance and body impedance measurements. Therefore, they are suited for bio-electric or biopotential measurement systems and for wearable applica-

tions too. These can even sense electrochemical toxic gases. They also contain some on-chip peripherals, including SPI serial input/output, wake-up timer, and interrupt controller.

The AD5940 can be used in conjunction with AD8232 or AD8233 to perform bioimpedance and ECG measurements. AD8232 and AD8233 are integrated signal conditioning blocks for ECG and other biopotential measurement applications.

For instance, when a bioimpedance measurement (for example, body composition, hydration, and electrodermal activity (EDA)) is required, AD8232/AD8233 are disconnected from the electrodes and AD5940/AD5941 are used. Likewise, when ECG measurements are needed, the opposite of this process happens.

These ICs have a structure different from the ones mentioned above. The AD5940 and AD5941 ICs contain two high-precision excitation loops. One is a low-power excitation loop and the second

consists of a 12-bit DAC referred to as the high-speed DAC. They have one measurement channel featuring a multichannel successive approximation register (SAR), ADC with input buffers, a built-in anti-alias filter, and a programmable gain amplifier (PGA). The measurement channel enables a wide capability of measurements of a sensor under test.

AFEs are not limited to medical/fitness applications. They can be used in any application that involves analogue sensing. Many IoT applications involve sensors being connected to a microcontroller and many of these sensors are analogue in nature. However, microcontrollers like Arduino Uno do not have a good analogue resolution and sensitivity. In such a scenario, one can consider interfacing AFEs with microcontrollers. **EFY**

This feature is inspired by an article written by Vladyslav Kozlov (Team Lead, Product Development Engineering), at Dialog Semiconductor, a Renesas company. The co-author, Aaryaa Padhye-gurjar, is an Industry 4.0 enthusiast with a keen interest in innovation and research.